ABES Engineering College, Ghaziabad

Department of Computer Science & Engineering

2nd Year (IVth Sem) Session 2025-26

CO1: Implement basic java programs using command line arguments.

CO2: Implement object-oriented programming concepts in java using Eclipse IDE.

CO3: Implement java programs for Interface and Packages.

CO4: Implement Java programs for Exception handling and Multithreading.

CO5: Develop industry-oriented Applications using spring boot and Spring framework.

Object Oriented Programming with Java Lab (BCS-452)

S.No.	Name of Practical	CO	K Level	Software's/ Tool's Required	AKTU Syllabus / Beyond Syllabus
1	Write a program to print addition of 2 matrices in java.	CO1	K3	VS Code	AKTU Syllabus
2	Write a program in java which creates the variable size array (Jagged Array) and print all the values using loop statement.	CO1	K3	VS Code	AKTU Syllabus
3	WAP that takes input from user through command line argument and then prints whether a number is prime or not.	CO1	K3	VS Code	AKTU Syllabus
4	Write a program to enter number through command line and check whether it is palindrome or not.	CO1	K3	VS Code	AKTU Syllabus
5	Write a Java program to create a Student class, initialize student details using a parameterized constructor, and display the student information using class objects and member functions.	CO2	K3	VS Code	AKTU Syllabus
6	Write a Java program to create a BankAccount class using encapsulation (private data members) and perform operations such as deposit, withdraw, and display balance using menu-driven choices.	CO2	K3	VS Code	AKTU Syllabus
7	Write a Java program to demonstrate Compile-time Polymorphism (Method Overloading) by creating a class Calculator that performs addition of integers, floating values, and three numbers using overloaded methods.	CO2	K3	VS Code	AKTU Syllabus
8	Write a Java program to demonstrate Single Inheritance by creating a Person class and deriving a Student class from it, and display the details of the student using inherited properties and methods.	CO2	K3	VS Code	AKTU Syllabus
9	Write a Java program to demonstrate Runtime Polymorphism (Method Overriding) by creating a base class Shape and derived classes Circle and Rectangle to override the area() method and display the area of each shape.	CO2	K3	VS Code	AKTU Syllabus
10	Write a Java program to justify that Java does not support multiple inheritance using classes, but multiple inheritance can be achieved using interfaces by implementing more than one interface in a single class.	CO2	K3	VS Code	AKTU Syllabus
11	Write a Java program to implement user defined exception handling for negative amount entered.	CO4	K3	VS Code	AKTU Syllabus
12	Write a Java program which creates two threads, "Even" thread and "Odd" thread and prints even numbers using Even Thread after every 2 seconds and odd numbers using Odd Thread after every 5 seconds.	CO4	K3	VS Code	AKTU Syllabus
13	Write a Java program to demonstrate exception handling using multiple catch blocks and finally block.	CO4	K3	VS Code	AKTU Syllabus
14	Write a Java program to demonstrate the use of throw and throws keyword to handle user input validation.	CO4	K3	VS Code	AKTU Syllabus
15	Write a Java program to create two threads using the Runnable interface and execute them simultaneously.	CO4	K3	VS Code	AKTU Syllabus
16	Create a package named Mathematics and add a class Matrix with methods to add and subtract matrices. Write a Java program importing the Mathematics package and use the classes defined in it.	CO3	K3	VS Code	AKTU Syllabus
17	Write a program in Java to take input from user by using all the following methods: 1. Command Line Arguments 2. DataInputStream Class 3. BufferedReader Class 4. Scanner Class 5. Console Class	CO3	K3	VS Code	AKTU Syllabus
18	Write a Java program to read username and password using Console class in such a way that the password is hidden while typing and then display the entered username and password.	CO3	K3	Notepad	AKTU Syllabus
19	Write a Spring Boot program to create an Employee Management System that performs CRUD operations (Create, Read, Update, Delete) using RESTful APIs.	CO5	K3	Eclipse/ Intellij	AKTU Syllabus
20	Write a Spring Boot program to create an Employee Management System that performs CRUD operations (Create, Read, Update, Delete) using RESTful APIs.	CO5	K3	Eclipse/ Intellij	AKTU Syllabus
21	Write a Spring Boot program to create and test a frontend web application for Employee Management System using Thymeleaf templates to Add and View Employees.	CO5	K3	Eclipse/ Intellij	AKTU Syllabus
22	Write a Spring Boot program to implement CRUD operations using REST API for Employee Management System using ArrayList / Map (in-memory storage).	CO5	K3	Eclipse/ Intellij	Beyond Syllabus

23	Write a Spring Boot application to develop a Book Store Management System using CRUD operations with REST API and in-memory storage using ArrayList. The application should allow the user to display all books, add a new book, update the price of an existing book using Book ID, and delete a book using Book ID through a Bootstrap-based web interface	CO5	K3	Eclipse/ Intellij	Beyond Syllabus
24	Write a Spring Boot application to develop a Book Store Management System using H2 Database and Spring Data JPA. The application should perform CRUD operations such as Add Book, View All Books, Update Book Price using Book ID, and Delete Book using Book ID through a Bootstrap-based web interface using Thymeleaf templates.	CO5	K3	Eclipse/ Intellij	Beyond Syllabus

HoD CSE